

2023–2024 学年期末真题

计算理论 · 回忆卷整理 (题目卷)

题型: 判断 · 不定项选择 · 大题

形式: 闭卷参考

整理用途: 课程资料复习

版本: 排版优化 · 仅题目

一、判断题

- The set of all regular languages over the alphabet $\{a, b\}$ is countable.
- Let $\Sigma = \emptyset$, some language over Σ is not empty.
- Let L_1, L_2 and L_3 be some recursively enumerable languages, then $L_1 L_2 \cap L_3^*$ must be recursively enumerable.
- Let L_1 be a regular language, L_3 be a context-free language. For any language $L_2 \subseteq L_3 - L_1$, although L_2 may not be regular, it must be context-free.
- Let L_1 and L_2 be some recursively enumerable languages, then $L_1 \oplus L_2$ must be recursively enumerable.
其中 $A \oplus B = (A \cup B) - (A \cap B)$.
- Let w be some string over $(\Sigma - \{\triangleright, \sqcup\})$. The following Turing machine transforms $\sqcup w \sqcup$ to $\sqcup w \sqcup$.
原卷附有具体 TM 状态转移图; 作答时按卷面图示理解即可.
- The set of all numerical functions is uncountable, but the set of all μ -recursive functions is countable.
- Let A be some language. Suppose that $A \leq \overline{A}$, it is impossible that A is not recursive but recursively enumerable.

二、不定项选择题

1. Choose the languages that may not be regular:

- A. the union of an infinite number of regular languages
- B. $\{w : w \in \{a, b\}^* \wedge w \neq w^R\}$
- C. $\{a^i b^j c^k : i + j \not\equiv k \pmod{2023}\}$
- D. $\{wvw^R : w \in \{a, b\}^+, v \in \{a, b\}^*\}$ (其中 $\{a, b\}^+ = \{a, b\}^* - \{e\}$)

2. L_1, L_2 are context-free languages. Which of the following languages must be context-free?

- A. $L_1 \cap L_2$
- B. $\overline{L_1}$
- C. $L_1 \oplus L_2$
- D. $L_1^* L_2^*$

3. Choose the languages that may not be context-free.

- A. $\{a^m b^n c^p : m = n \vee n = p \vee p = m\}$
- B. $\{a^m b^n c^p : m \neq n \vee n \neq p \vee p \neq m\}$
- C. $L - R$, 其中 L 是 CFL, R 是正则
- D. $R - L$, 其中 L 是 CFL, R 是正则

4. Choose true statements.

- A. Any recursive language must be recursively enumerable
- B. The class of recursively enumerable languages is closed under intersection
- C. If a language satisfies the pumping theorem, it must be regular
- D. If a language L is recursively enumerable, so is $\overline{L}$

三、大题

大题 1

判断以下两个语言是否正则并证明:

- 1. $A = \{1^k y : y \in \{0, 1\}^*, y \text{ contains at most } k \text{ 1's, } k \geq 1\}$
- 2. $B = \{1^k y : y \in \{0, 1\}^*, y \text{ contains at least } k \text{ 1's, } k \geq 1\}$

大题 2

Design a context-free grammar that generates

$$\{ww^R c a^n b^m : w \in \{a, b\}^*, n > m, n \text{ is even}, m \text{ is odd}\}$$

over $\{a, b, c\}$.

大题 3

Let RE be the set of all recursively enumerable languages. Let $\text{co-RE} = \{A : \overline{A} \text{ is recursively enumerable}\}$.

Consider the Venn diagram of RE 与 co-RE . Which zone do $H, H_1, \overline{H}, \overline{H_1}$ belong? What about P and NP ? (Assume $P \neq NP$.)

其中:

- $H = \{\langle M \rangle \langle w \rangle : M \text{ halts on } w\}$
- $H_1 = \{\langle M \rangle : M \text{ halts on } \langle M \rangle\}$

大题 4

Let $M = (K, \Sigma, \delta, s, \{h\})$ be a standard Turing machine. Assume that whenever M halts, it's always in configuration $(h, \triangleright \sqcup)$. Describe how to construct a grammar that generates $L(M)$.

大题 5

Consider the following languages. Are they recursive, RE but not recursive, or not RE? Justify.

1. $A = \{\langle M \rangle : M \text{ is the only TM that semidecides } L(M)\}$
2. $B = \{\langle M \rangle : M \text{ is a TM, } L(M) \text{ is finite}\}$